

Optimized Table Tokenization for Table Structure Recognition

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Abstract. Extracting tables from documents is a crucial task in any document conversion pipeline. Recently, transformer-based models have demonstrated that table-structure can be recognized with impressive accuracy using Image-to-Markup-Sequence (Im2Seq) approaches. Taking only the image of a table, such models predict a sequence of tokens (e.g. in HTML, LaTeX) which represent the structure of the table. Since the token representation of the table structure has a significant impact on the accuracy and run-time performance of any Im2Seq model, we investigate in this paper how table-structure representation can be optimised. We propose a new, optimised table-structure language (OTSL) with a minimized vocabulary and specific rules. The benefits of OTSL are that it reduces the number of tokens to 5 (HTML needs 28+) and shortens the sequence length to half of HTML on average. Consequently, model accuracy improves significantly, inference time is halved compared to HTML-based models, and the predicted table structures are always syntactically correct. This in turn eliminates most post-processing needs. Popular table structure data-sets will be published in OTSL format to the community.

Keywords: Table Structure Recognition · Data Representation · Transformers · Optimization.

1 Introduction

Tables are ubiquitous in documents such as scientific papers, patents, reports, manuals, specification sheets or marketing material. They often encode highly valuable information and therefore need to be extracted with high accuracy. Unfortunately, tables appear in documents in various sizes, styling and structure, making it difficult to recover their correct structure with simple analytical methods. Therefore, accurate table extraction is achieved these days with machine-learning based methods.

In modern document understanding systems [1,15], table extraction is typically a two-step process. Firstly, every table on a page is located with a bounding box, and secondly, their logical row and column structure is recognized. As of

While the majority of research in TSR is currently focused on the development and application of novel neural model architectures, the table structure representation language (e.g. HTML in PubTabNet and FinTabNet) is usually adopted *as is* for the sequence tokenization in Im2Seq models. In this paper, we aim for the opposite and investigate the impact of the table structure representation language with an otherwise unmodified Im2Seq transformer-based architecture. Since the current state-of-the-art Im2Seq model is TableFormer [9], we select this model to perform our experiments.

The main contribution of this paper is the introduction of a new optimised table structure language (OTSL), specifically designed to describe table-structure in an compact and structured way for Im2Seq models. OTSL has a number of key features, which make it very attractive to use in Im2Seq models. Specifically, compared to other languages such as HTML, OTSL has a minimized vocabulary which yields short sequence length, strong inherent structure (e.g. strict rectangular layout) and a strict syntax with rules that only look backwards. The latter allows for syntax validation during inference and ensures a syntactically correct table-structure. These OTSL features are illustrated in Figure 1, in comparison to HTML.

The paper is structured as follows. In section 2, we give an overview of the latest developments in table-structure reconstruction. In section 3 we review the current HTML table encoding (popularised by PubTabNet and FinTabNet) and discuss its flaws. Subsequently, we introduce OTSL in section 4, which includes the language definition, syntax rules and error-correction procedures. In section 5, we apply OTSL on the TableFormer architecture, compare it to TableFormer models trained on HTML and ultimately demonstrate the advantages of using OTSL. Finally, in section 6 we conclude our work and outline next potential steps.

2 Related Work

Approaches to formalize the logical structure and layout of tables in electronic documents date back more than two decades [16]. In the recent past, a wide variety of computer vision methods have been explored to tackle the problem of table structure recognition, i.e. the correct identification of columns, rows and spanning cells in a given table. Broadly speaking, the current deep-learning based approaches fall into three categories: object detection (OD) methods, Graph-Neural-Network (GNN) methods and Image-to-Markup-Sequence (Im2Seq) methods. Object-detection based methods [11,12,13,14,21] rely on table-structure annotation using (overlapping) bounding boxes for training, and produce bounding-box predictions to define table cells, rows, and columns on a table image. Graph Neural Network (GNN) based methods [3,6,17,18], as the name suggests, represent tables as graph structures. The graph nodes represent the content of each table cell, an embedding vector from the table image, or geometric coordinates of the table cell. The edges of the graph define the relationship between the nodes, e.g. if they belong to the same column, row, or table cell.

Other work [20] aims at predicting a grid for each table and deciding which cells must be merged using an attention network. Im2Seq methods cast the problem as a sequence generation task [4,5,9,22], and therefore need an internal table-structure representation language, which is often implemented with standard markup languages (e.g. HTML, LaTeX, Markdown). In theory, Im2Seq methods have a natural advantage over the OD and GNN methods by virtue of directly predicting the table-structure. As such, no post-processing or rules are needed in order to obtain the table-structure, which is necessary with OD and GNN approaches. In practice, this is not entirely true, because a predicted sequence of table-structure markup does not necessarily have to be syntactically correct. Hence, depending on the quality of the predicted sequence, some post-processing needs to be performed to ensure a syntactically valid (let alone correct) sequence.

Within the Im2Seq method, we find several popular models, namely the encoder-dual-decoder model (EDD) [22], TableFormer [9], Tabsplitter[2] and Ye et. al. [19]. EDD uses two consecutive long short-term memory (LSTM) decoders to predict a table in HTML representation. The *tag decoder* predicts a sequence of HTML tags. For each decoded table cell (`<td>`), the attention is passed to the *cell decoder* to predict the content with an embedded OCR approach. The latter makes it susceptible to transcription errors in the cell content of the table. TableFormer address this reliance on OCR and uses two transformer decoders for HTML structure and cell bounding box prediction in an end-to-end architecture. The predicted cell bounding box is then used to extract text tokens from an originating (digital) PDF page, circumventing any need for OCR. TabSplitter [2] proposes a compact double-matrix representation of table rows and columns to do error detection and error correction of HTML structure sequences based on predictions from [19]. This compact double-matrix representation can not be used directly by the Im2seq model training, so the model uses HTML as an intermediate form. Chi et. al. [4] introduce a data set and a baseline method using bidirectional LSTMs to predict LaTeX code. Kayal [5] introduces Gated ResNet transformers to predict LaTeX code, and a separate OCR module to extract content.

Im2Seq approaches have shown to be well-suited for the TSR task and allow a full end-to-end network design that can output the final table structure without pre- or post-processing logic. Furthermore, Im2Seq models have demonstrated to deliver state-of-the-art prediction accuracy [9]. This motivated the authors to investigate if the performance (both in accuracy and inference time) can be further improved by optimising the table structure representation language. We believe this is a necessary step before further improving neural network architectures for this task.

3 Problem Statement

All known Im2Seq based models for TSR fundamentally work in similar ways. Given an image of a table, the Im2Seq model predicts the structure of the table by generating a sequence of tokens. These tokens originate from a finite vocab-

generation. Implicitly, this also means that Im2Seq models need to learn these complex syntax rules, simply to deliver valid output.

In practice, we observe two major issues with prediction quality when training Im2Seq models on HTML table structure generation from images. On the one hand, we find that on large tables, the visual attention of the model often starts to drift and is not accurately moving forward cell by cell anymore. This manifests itself in either in an increasing *location drift* for proposed table-cells in later rows on the same column or even complete loss of vertical alignment, as illustrated in Figure 5. Addressing this with post-processing is partially possible, but clearly undesired. On the other hand, we find many instances of predictions with structural inconsistencies or plain invalid HTML output, as shown in Figure 6, which are nearly impossible to properly correct. Both problems seriously impact the TSR model performance, since they reflect not only in the task of pure structure recognition but also in the equally crucial recognition or matching of table cell content.

4 Optimised Table Structure Language

To mitigate the issues with HTML in Im2Seq-based TSR models laid out before, we propose here our Optimised Table Structure Language (OTSL). OTSL is designed to express table structure with a minimized vocabulary and a simple set of rules, which are both significantly reduced compared to HTML. At the same time, OTSL enables easy error detection and correction during sequence generation. We further demonstrate how the compact structure representation and minimized sequence length improves prediction accuracy and inference time in the TableFormer architecture.

4.1 Language Definition

In Figure 3, we illustrate how the OTSL is defined. In essence, the OTSL defines only 5 tokens that directly describe a tabular structure based on an atomic 2D grid.

The OTSL vocabulary is comprised of the following tokens:

- "C" cell - *a new table cell* that either has or does not have cell content
- "L" cell - *left-looking cell*, merging with the left neighbor cell to create a span
- "U" cell - *up-looking cell*, merging with the upper neighbor cell to create a span
- "X" cell - *cross cell*, to merge with both left and upper neighbor cells
- "NL" - *new-line*, switch to the next row.

A notable attribute of OTSL is that it has the capability of achieving lossless conversion to HTML.

6 Conclusion

We demonstrated that representing tables in HTML for the task of table structure recognition with Im2Seq models is ill-suited and has serious limitations. Furthermore, we presented in this paper an Optimized Table Structure Language (OTSL) which, when compared to commonly used general purpose languages, has several key benefits.

First and foremost, given the same network configuration, inference time for a table-structure prediction is about 2 times faster compared to the conventional HTML approach. This is primarily owed to the shorter sequence length of the OTSL representation. Additional performance benefits can be obtained with HPO (hyper parameter optimization). As we demonstrate in our experiments, models trained on OTSL can be significantly smaller, e.g. by reducing the number of encoder and decoder layers, while preserving comparatively good prediction quality. This can further improve inference performance, yielding 5-6 times faster inference speed in OTSL with prediction quality comparable to models trained on HTML (see Table 1).

Secondly, OTSL has more inherent structure and a significantly restricted vocabulary size. This allows autoregressive models to perform better in the TED metric, but especially with regards to prediction accuracy of the table-cell bounding boxes (see Table 2). As shown in Figure 5, we observe that the OTSL drastically reduces the drift for table cell bounding boxes at high row count and in sparse tables. This leads to more accurate predictions and a significant reduction in post-processing complexity, which is an undesired necessity in HTML-based Im2Seq models. Significant novelty lies in OTSL syntactical rules, which are few, simple and always backwards looking. Each new token can be validated only by analyzing the sequence of previous tokens, without requiring the entire sequence to detect mistakes. This in return allows to perform structural error detection and correction on-the-fly during sequence generation.

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